

CIP-B101 Basic Computer Skills for Digital Forensics

Lab 1C: Windows Command Line for Digital Forensics Basics

Student Practical Laboratory Manual

Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Title	Windows Command Line for Digital Forensics Basics
Recommended Environment	Windows 10/11 host or Windows VM with Command Prompt access
Estimated Duration	2-3 hours
Submission Format	One PDF report plus the created .bat files and command-output evidence files

Important preparation note: Before attempting this lab, trainees must carefully go through the Windows Command Line Tutorial slide material because the slide is a guided walkthrough. This lab converts that walkthrough into a structured forensic practical exercise with screenshots, evidence capture, and reporting requirements.

Lab safety: Perform this activity only on your own computer or an approved training VM. Do not collect information from systems you do not own or do not have permission to examine. Do not delete important files outside the lab workspace.

1. Lab Scenario

You are a trainee digital forensic analyst assigned to perform a basic Windows command-line inspection on a training workstation. Your task is to document the Windows file-system layout, create and manage evidence folders, collect basic system and network information, search for evidence text inside files, and build a small batch file that can quickly collect repeatable system information.

The goal is not to investigate a real incident. The goal is to build command-line discipline, evidence documentation habits, and repeatable steps that are useful in later digital forensic work.

2. Learning Outcomes

- Explain absolute and relative paths in Windows.
- Navigate directories using Command Prompt.
- Create, copy, rename, delete, and verify files and folders using commands.
- Collect basic host information, network configuration, MAC addresses, disks, CPU, and process information.
- Search text files using command-line search commands.
- Create and execute a batch file for repeatable system-information collection.
- Modify PATH so a batch file can be executed from any directory.
- Prepare a clear forensic-style lab report with screenshots and command outputs.

3. What Students Need for This Lab

Requirement	Details
Computer/VM	Windows 10 or Windows 11 computer, or Windows virtual machine.
Access level	Standard user is enough for most tasks. Administrator access may be needed for

	changing system PATH permanently.
Tools	Command Prompt, Notepad, optional Windows Terminal, and a PDF editor/Word processor for the report.
Internet	Optional only for opening the Microsoft Windows system requirements link used in the sample batch file.
Lab workspace	C:\Users\Public\CIP-B101_Lab1C or another approved lab-only folder.
Evidence report	A final PDF report with screenshots, command outputs, and short explanations.

4. Reference URL Extracted from the Slide

The Windows Command Line Tutorial slide includes the following Microsoft Windows system requirements reference URL inside the sample batch file. Students may use it only as a reference; the lab can still be completed without opening it:

<https://support.microsoft.com/en-us/windows/windows-10-system-requirements-6d4e9a79-66bf-7950-467c-795cf0386715>

5. Before You Start: Create Your Evidence Folder

Open Command Prompt and create a dedicated lab workspace. Replace FIRSTNAME_LASTNAME with your real name or registration name.

```
cd \Users\Public
mkdir CIP-B101_Lab1C_FIRSTNAME_LASTNAME
cd CIP-B101_Lab1C_FIRSTNAME_LASTNAME
mkdir evidence screenshots outputs scripts workspace workspace2 workspace3 workspace4
dir
```

Take Screenshot 1 showing the created lab directory and subfolders.

Task 1: Understand Windows Paths and Navigation

In this task, you will practice absolute paths, relative paths, root directories, and parent-directory navigation. This is important because forensic analysts must correctly identify where evidence is located.

1.1 Display current location

```
cd
echo Current directory is: %cd%
```

1.2 Move to the Windows root directory

```
cd \
dir
```

1.3 Move to Users and then Public

```
cd \Users
dir
cd Public
dir
```

1.4 Move back using a relative path

```
cd ..
cd Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME
```

1.5 Display the folder structure

tree

Evidence required: Screenshot showing cd, dir, and tree outputs. In your report, define absolute path and relative path in your own words.

Task 2: Basic File and Folder Creation

This task teaches basic command-line file creation and verification. You will create a text file that represents a simple evidence note.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\workspace
echo hello world > my_first_file.txt
type my_first_file.txt
echo Student Name: FIRSTNAME LASTNAME > evidence_note.txt
echo Course: CIP-B101 Lab 1C >> evidence_note.txt
echo Purpose: Windows command line forensic basics >> evidence_note.txt
type evidence_note.txt
dir
```

Evidence required: Screenshot showing both files and their contents. Explain the difference between > and >> in command output redirection.

Task 3: Copy, Duplicate, Rename, and Delete Files Safely

Forensic work requires careful handling of files. In this exercise, you will copy files and verify that the copies exist before deleting test files.

3.1 Copy a file to another folder

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\workspace
copy evidence_note.txt ..\workspace2\evidence_note_copy.txt
dir ..\workspace2
```

3.2 Copy a whole folder recursively

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME
xcopy workspace workspace3\workspace_backup /E /I /H
tree workspace3
```

3.3 Rename a copied file

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\workspace2
ren evidence_note_copy.txt renamed_evidence_note.txt
dir
```

3.4 Delete only a test file

```
echo temporary test file > delete_me.txt
dir
del delete_me.txt
dir
```

3.5 Remove an empty folder

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME
mkdir empty_test_folder
rmdir empty_test_folder
```

```
dir
```

Evidence required: Screenshot of copied folder, renamed file, and successful deletion of only the temporary test file. Do not delete your evidence folders.

Task 4: Understand How Windows Finds Applications

Windows searches executable files using defined paths. This is important because attackers and malware may abuse PATH or place malicious executables in searchable directories.

```
echo %PATH%  
where notepad  
notepad  
vol C:  
cls
```

Evidence required: Screenshot showing the PATH variable and where notepad is located. In your report, explain why PATH matters during malware triage and forensic review.

Task 5: Collect Windows System Information

This task collects host details that are commonly included in a basic forensic triage note.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\outputs  
systeminfo > systeminfo_full.txt  
systeminfo | find "OS Name"  
systeminfo | find "OS Version"  
systeminfo | find "System Type"  
systeminfo | find "Total Physical Memory"  
wmic cpu get name > cpu_info.txt  
wmic diskdrive get name,model,size > disk_info.txt  
type cpu_info.txt  
type disk_info.txt
```

Evidence required: Screenshots showing OS name/version, system type, physical memory, CPU name, and disk model/size. Also include systeminfo_full.txt in your submitted ZIP if requested by the instructor.

Task 6: Collect Basic Network Information

This task records IP configuration, default gateway information, MAC address information, and basic connectivity checks.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\outputs  
ipconfig /all > ipconfig_all.txt  
ipconfig | findstr IPv4  
ipconfig | findstr Gateway  
wmic nic get macaddress,description > mac_addresses.txt  
type mac_addresses.txt  
ping 127.0.0.1  
ping 8.8.8.8  
ping google.com
```

Evidence required: Screenshot of IPv4 address, gateway, MAC address output, and at least one successful ping. If internet ping fails, explain the likely reason instead of hiding the failure.

Task 7: Search for Evidence Text in Files

You will create a small text evidence file and then search it using Windows command-line search tools. This prepares you for later file-content searching in forensic work.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\evidence
echo User: FIRSTNAME LASTNAME > mini_case_log.txt
echo Email: trainee@example.com >> mini_case_log.txt
echo IP: 192.168.56.10 >> mini_case_log.txt
echo Keyword: suspicious_login >> mini_case_log.txt
type mini_case_log.txt
findstr /i "email" mini_case_log.txt
findstr /i "suspicious" mini_case_log.txt
findstr /n /i "ip" mini_case_log.txt
```

Evidence required: Screenshot showing the created mini_case_log.txt and the search results. Explain the function of /i and /n in findstr.

Task 8: Check File Owner and Directory Information

The slide introduces file ownership review. In forensic documentation, ownership and access context can be useful when reviewing a local system.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\evidence
dir /q
icacls mini_case_log.txt
```

Evidence required: Screenshot showing owner information and permission output for mini_case_log.txt. Briefly describe who owns the file on your system.

Task 9: List Processes and Identify Browser Processes

Processes are running instances of programs. During basic incident triage, analysts often collect process information to identify suspicious or unusual activity.

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\outputs
wmic process list brief > process_list.txt
type process_list.txt | more
wmic process where "name like '%chrome%'" get name,processid,commandline
wmic process where "name like '%msedge%'" get name,processid,commandline
```

Evidence required: Screenshot showing general process listing and either Chrome or Edge process search. If no browser process appears, open the browser and run the command again.

Task 10: Create and Execute a Batch File

Batch files allow analysts to run repeatable commands. In this task, you will create a simple Hello World batch file and a forensic-style system-information batch file.

10.1 Create hello_world.bat

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\scripts
notepad hello_world.bat
```

Paste the following content into Notepad, save the file, and close Notepad:

```
@ECHO OFF
```

ECHO Hello World!

PAUSE

Run it:

hello_world.bat

10.2 Create sys_info.bat

In the same scripts folder, run:

notepad sys_info.bat

Paste the following content:

```
@ECHO OFF
:: CIP-B101 Lab 1C - Windows system information collection
TITLE CIP-B101 System Info Collection
ECHO Please wait... Checking system information.
ECHO ===== > ..\outputs\sys_info_report.txt
ECHO WINDOWS INFO >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
systeminfo | find "OS Name" >> ..\outputs\sys_info_report.txt
systeminfo | find "OS Version" >> ..\outputs\sys_info_report.txt
systeminfo | find "System Type" >> ..\outputs\sys_info_report.txt
ECHO. >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
ECHO HARDWARE INFO >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
systeminfo | find "Total Physical Memory" >> ..\outputs\sys_info_report.txt
wmic cpu get name >> ..\outputs\sys_info_report.txt
wmic diskdrive get name,model,size >> ..\outputs\sys_info_report.txt
ECHO. >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
ECHO NETWORK INFO >> ..\outputs\sys_info_report.txt
ECHO ===== >> ..\outputs\sys_info_report.txt
ipconfig | findstr IPv4 >> ..\outputs\sys_info_report.txt
wmic nic get macaddress,description >> ..\outputs\sys_info_report.txt
START https://support.microsoft.com/en-us/windows/windows-10-system-requirements-6d4e9a79-66bf-7950-467c-795cf0386715
ECHO Collection complete. See outputs\sys_info_report.txt
PAUSE
```

Run the batch file and verify the output file:

sys_info.bat

type ..\outputs\sys_info_report.txt

Evidence required: Screenshot showing sys_info.bat execution and the generated sys_info_report.txt output.

Task 11: Make sys_info.bat Callable from Any Folder

The original slide assignment asks students to make sys_info.bat callable from any folder. This means Windows must be able to locate the batch file through PATH.

Create a new folder named workspace4 and copy sys_info.bat into it:

```
cd \Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME
```

```
copy scripts\sys_info.bat workspace4\sys_info.bat
```

```
dir workspace4
```

Option A - temporary PATH update for the current Command Prompt session only:

```
set PATH=%PATH%;C:\Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\workspace4
```

```
cd \
```

```
sys_info.bat
```

Option B - permanent PATH update for the user account. Use only if your instructor permits it:

```
setx PATH "%PATH%;C:\Users\Public\CIP-B101_Lab1C_FIRSTNAME_LASTNAME\workspace4"
```

Evidence required: Screenshot showing sys_info.bat running successfully from a folder other than its own folder. In the report, state whether you used temporary or permanent PATH update.

Task 12: Mini Forensic Interpretation Questions

1. What is the difference between an absolute path and a relative path? Give one example of each from your lab.
2. Why should analysts create a separate lab/evidence folder before running commands?
3. Which command showed your IP address and which command showed your MAC address?
4. Why is a batch file useful in forensic triage?
5. Why must analysts document failed commands or failed pings instead of hiding them?
6. What risk can occur if a suspicious executable is placed in a directory listed in PATH?

6. Screenshot Checklist

- Lab workspace and folders created.
- Path navigation using cd, dir, and tree.
- Created text files and displayed file contents using type.
- Copy, folder backup, rename, delete, and rmdir outputs.
- PATH variable and where notepad output.
- systeminfo and WMIC CPU/disk outputs.
- ipconfig, MAC address, and ping outputs.
- findstr search results from mini_case_log.txt.
- dir /q or icacls ownership/permission output.
- process list and browser process search.
- hello_world.bat execution.
- sys_info.bat execution and sys_info_report.txt output.
- sys_info.bat running from another folder after PATH update.

7. Submission Deliverables

Deliverable	Requirement
PDF Lab Report	Must include title page, all screenshots, command explanations, and answers to the mini forensic questions.
Batch Files	Submit hello_world.bat and sys_info.bat.
Output Files	Submit systeminfo_full.txt, cpu_info.txt, disk_info.txt, ipconfig_all.txt, mac_addresses.txt, process_list.txt, and sys_info_report.txt if requested.
Folder Naming	CIP-B101_Lab1C_FIRSTNAME_LASTNAME.zip
Academic Integrity	All screenshots must be from the student's own machine or approved VM.

8. Suggested Report Structure

- Cover page: student name, registration number, course code, lab title, date.
- Objective: short explanation of what the lab is about.
- Environment: Windows version, device/VM type, user account type.
- Methodology: command groups used and why they were used.
- Evidence screenshots: label each screenshot and explain what it proves.
- Collected outputs: summarize important system/network information found.
- Batch file section: explain what sys_info.bat collects and show it running from another folder.
- Mini forensic interpretation answers.
- Conclusion: what you learned and how the command line supports digital forensics.

9. Marking Rubric - 100 Marks

Assessment Area	Marks
Preparation and safe workspace setup	10
Windows path navigation and file-system understanding	10
File/folder creation, copy, rename, and deletion tasks	15
System, disk, CPU, and process information collection	15
Network, MAC address, and connectivity evidence	10
Text search and ownership/permission review	10
Batch file creation and PATH execution task	15
Screenshots, report quality, explanations, and professionalism	15

10. Common Mistakes to Avoid

- Running commands in the wrong folder and then losing evidence files.
- Using delete commands outside the lab folder.
- Submitting screenshots without explaining what they prove.
- Changing PATH permanently without permission from the instructor.
- Copying another student's screenshots or batch files.
- Failing to include the generated output files in the final submission.

End of Lab 1C. After completing this practical, students should be comfortable using the Windows command line to navigate the file system, collect basic system and network information, create repeatable batch scripts, and document evidence in a forensic-style report.